

Sri Lanka Institute of Information Technology



Database Systems – SE3060 Laboratory Worksheet 1

1. Create the relational tables in Oracle with suitable primary and foreign keys, and insert the sample data

```
CREATE TABLE client (  
  clno CHAR(3),  
  name VARCHAR(12),  
  address VARCHAR(30),  
  CONSTRAINT client_pk PRIMARY KEY(clno)  
);
```

```
CREATE TABLE stock (  
  company CHAR(7),  
  price NUMBER(6,2),  
  dividend NUMBER(4,2),  
  eps NUMBER(4,2),  
  CONSTRAINT stock_pk PRIMARY KEY (company)  
);
```

```
CREATE TABLE trading(  
  company CHAR(7),  
  exchange VARCHAR(12),  
  CONSTRAINT trading_pk PRIMARY KEY(company,exchange),  
  CONSTRAINT trading_fk FOREIGN KEY(company) REFERENCES stock  
);
```

```
CREATE TABLE purchase(  
  clno CHAR(3),  
  company CHAR(7),  
  pdate DATE,  
  qty NUMBER(6),  
  price NUMBER(6,2),  
  CONSTRAINT purchase_pk PRIMARY KEY(clno,company,pdate),  
  CONSTRAINT purchase_fk1 FOREIGN KEY(clno) REFERENCES client,  
  CONSTRAINT purchase_fk2 FOREIGN KEY(company) REFERENCES stock  
);
```

```
INSERT INTO client VALUES('c01', 'John Smith', '3 East Av Bentley WA 6102');  
INSERT INTO client VALUES('c02', 'Jill Brody', '42 Bent St Perth WA 6001');
```

```
INSERT INTO stock VALUES('BHP', 10.50, 1.50 , 3.20);  
INSERT INTO stock VALUES('IBM', 70.00, 4.25, 10.00);  
INSERT INTO stock VALUES('INTEL', 76.50 , 5.00 , 12.40);  
INSERT INTO stock VALUES('FORD', 40.00, 2.00, 8.50);  
INSERT INTO stock VALUES('GM', 60.00 , 2.50 , 9.20);  
INSERT INTO stock VALUES('INFOSYS', 45.00 , 3.00, 7.80);
```

```
INSERT INTO trading VALUES ('BHP', 'Sydney');  
INSERT INTO trading VALUES ('BHP', 'New York');  
INSERT INTO trading VALUES ('IBM', 'New York');  
INSERT INTO trading VALUES ('IBM', 'London');  
INSERT INTO trading VALUES ('IBM', 'Tokyo');  
INSERT INTO trading VALUES ('INTEL', 'New York');  
INSERT INTO trading VALUES ('INTEL', 'London');  
INSERT INTO trading VALUES ('FORD', 'New York');  
INSERT INTO trading VALUES ('GM', 'New York');  
INSERT INTO trading VALUES ('INFOSYS', 'New York');
```

```
INSERT INTO purchase VALUES ('c01', 'BHP', TO_DATE('2001-10-02', 'YYYY-MM-DD'), 1000, 12.00);  
INSERT INTO purchase VALUES ('c01', 'BHP', TO_DATE('2002-06-08', 'YYYY-MM-DD'), 2000, 10.50);  
INSERT INTO purchase VALUES ('c01', 'IBM', TO_DATE('2000-02-12', 'YYYY-MM-DD'), 500, 58.00);  
INSERT INTO purchase VALUES ('c01', 'IBM', TO_DATE('2001-04-10', 'YYYY-MM-DD'), 1200, 65.00);  
INSERT INTO purchase VALUES ('c01', 'INFOSYS', TO_DATE('2001-08-11', 'YYYY-MM-DD'), 1000, 64.00);  
INSERT INTO purchase VALUES ('c02', 'INTEL', TO_DATE('2000-01-30', 'YYYY-MM-DD'), 300, 35.00);  
INSERT INTO purchase VALUES ('c02', 'INTEL', TO_DATE('2001-01-30', 'YYYY-MM-DD'), 400, 54.00);  
INSERT INTO purchase VALUES ('c02', 'INTEL', TO_DATE('2001-10-02', 'YYYY-MM-DD'), 200, 60.00);  
INSERT INTO purchase VALUES ('c02', 'FORD', TO_DATE('1999-10-05', 'YYYY-MM-DD'), 300, 40.00);
```

```
INSERT INTO purchase VALUES ('c02', 'GM', TO_DATE('2000-12-12', 'YYYY-MM-DD'), 500, 55.50);
```

2. Answer the queries given below, using the tables in your database.

i.

```
SELECT c.name, s.company, s.price, s.dividend, s.eps  
  
FROM purchase p  
  
JOIN client c ON p.clno = c.clno  
  
JOIN stock s ON s.company = p.company  
  
ORDER BY c.name;
```

ii.

```
SELECT  
  
c.name AS client_name,  
p.company AS stock_name,  
SUM(p.qty) AS total_shares,  
ROUND(SUM(p.qty * p.price) / SUM(p.qty), 2) AS avg_purchase_price  
FROM purchase p JOIN client c ON p.clno = c.clno  
  
GROUP BY c.name, p.company  
  
ORDER BY c.name, p.company;
```

iii.

```
SELECT  
  
c.name,  
p.company,  
SUM(p.qty) AS total_qty,  
ROUND(SUM(p.qty) * s.price, 2) AS current_value_shares  
FROM purchase p  
  
JOIN client c ON p.clno = c.clno
```

```
JOIN trading t ON p.company = t.company  
JOIN stock s ON p.company = s.company  
WHERE t.exchange = 'New York'  
GROUP BY c.name, p.company, s.price;
```

iv.

```
SELECT  
c.name,  
SUM(p.qty * p.price) AS total_purchase_value  
FROM client c  
JOIN purchase p ON c.clno = p.clno  
GROUP BY c.name;
```

v.

```
SELECT  
c.name,  
ROUND(SUM(p.qty * s.price) - SUM(p.qty * p.price), 2) AS book_profit_or_loss  
FROM client c  
JOIN purchase p ON c.clno = p.clno  
JOIN stock s ON p.company = s.company  
GROUP BY c.name  
ORDER BY c.name;
```